OceanSteel AI

Autonomous Maritime-to-Inland Decision Support & Total Landed Cost (TLC) Optir

Problem Statement ID: SIH262006 | Category: Software | Target: Ministry of Steel, SAIL, RINL

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HE REAL-WORLD INDUSTRY CRISIS

- **High Raw Material Logistics Share:** Raw material transport accounts for up to 40% of India's crude steel cost.
- **Heavy Coking Coal Dependence:** India imports 85%+ of high-grade metallurgical coking coal from Australia/South Africa.
- **Severe Port Congestion:** When giant bulk vessels arrive at East Coast ports (Paradip, Vizag), they face 5–10 day queues.
- **Massive Demurrage Losses:** Every idle day costs 20,000to35,000 USD (₹16–30 Lakhs/day) in foreign exchange fines.
- **Single Voyage Waste:** A single 5-to-7 day delay wastes ₹1.5 to ₹2.2 Crores per shipment on avoidable port waiting fines.
- **The Operational Silo:** Maritime tracking (AIS) and Indian Railways freight (FOIS/CRIS) do NOT talk to each other!

HE OCEANSTEEL AI SOLUTION

- **The Intermodal Bridge:** Connects live ship GPS (AIS) with Indian Railways freight train availability (FOIS).
- **Prescriptive (Not Just Descriptive):** Tools like MarineTraffic only show where a ship is; our system tells you what to do.
- **10–14 Day Advance Notice:** Calculates congestion while the ship is still in the Indian Ocean, long before it reaches India.
- **Smart Total Landed Cost:** Checks if burning a little fuel to divert to Dhamra saves millions in port waiting fines.
- **Strict Physical Constraints:** Enforces vessel water depth limits (draft) and checks that railway wagons are available.
- **Immediate 1-Click Execution:** Auto-generates the legal BIMCO Rerouting Addendum and books railway trains on CRIS.

D RESULTS ON BENCHMARK SHIP (MV STEEL H

₹2.26 CRORE ($272k)	**8.3 DAYS EARLIER**	**6.7 DAYS AVOIDED**	**39 BOXN RAKES**
Net Financial Savings	Plant Delivery Ahead of Schedule	Waiting Time at Port Anchorage	Trains Synchronized via FOIS

BENCHMARK VOYAGE: MV STEEL HORIZON

Cargo: 150,000 MT Australian Premium Hard Coking Coal | Vessel Type: Capesize Bulk Carrier (17.5m Draft)

Origin: Gladstone, Queensland | Destination Plant: SAIL Rourkela Steel Plant (RSP), Odisha

AL & OPERATIONAL COMPARISON: PARADIP vs

Parameter	Nominated Port (Paradip)	Recommended Port (Dhamra)	Variance / Net De
Anchorage Waiting Queue	7.8 Days (Heavy congestion)	1.1 Days (Free mechanized berth)	**-6.7 Days Avoided**
Demurrage Penalty ($25k/day)	$195,000 USD (₹1.62 Crore)	$27,500 USD (₹22.8 Lakhs)	**-$167,500 USD Saved**
Extra Sea Bunker Fuel	$0 USD (Baseline)	+$4,297 USD (0.18 days steaming)	**+$4,297 USD (Minor fuel)**
Rail Distance to Rourkela	462 km (East Coast Railway)	418 km (South Eastern Railway)	**-44 km Shorter Rail Route**
Indian Railways Freight (Class 140)	$2,307,692 USD (₹19.20 Crore)	$2,181,490 USD (₹18.15 Crore)	**-$126,202 USD Saved (₹1.05 Cr)**
Port Handling & Wharfage Dues	630,000*USD*(4.20/MT)	667,500*USD*(4.45/MT)	**+$37,500 USD (Slightly higher)**
TOTAL LANDED COST (TLC)	**$5,027,855 USD (₹41.83 Crore)**	**$4,755,787 USD (₹39.57 Crore)**	**NET SAVINGS: -$272,068 USD (₹2.26 Cr**

PRESCRIPTIVE DECISION GATE FORMULA

HOW THE PYTHON ENGINE DECIDES TO DIVERT:

Δ Demurrage Saved > Δ Sea Fuel + Δ Inland Rail Freight + Δ Port Handling

1. Demurrage Saved (167,500)*completelydwarfsthetinyseafuelburn*(4,297 for 4.3 hours of extra steaming).
2. Rail Distance Bonus: Dhamra Port is 44 km closer by rail to Rourkela than Paradip, saving ₹70 per tonne on railway freight!
3. Water Depth Feasibility: Dhamra port has 18.5m draft (our ship needs 17.5m - perfectly safe!). Haldia was blocked (only 8.5m).
4. Critical Plant Runway: SAIL Rourkela had only 11 days of coal buffer. Dhamra route delivers 8.3 days earlier, avoiding plant shutdown.

HOW OCEANSTEEL AI WORKS BEHIND THE SCENES

A simple, elegant 4-module pipeline connecting ships, ports, railways, and steel mills

MODULE 1: GIS TACTICAL COMMAND MAP

- **Tech:** Leaflet.js + Esri Dark Canvas GIS
- **What it does:** Visualizes vessels sailing in the Indian Ocean, port berths, and coal sidings.
- **Key features:** Live vessel GPS coordinates, heading, and speed.
- **Port Status:** Color-coded port congestion beacons (Red: jammed, Green: ready).
- **Corridors:** Shows freight rail lines connecting ports to steel plants.

MODULE 2: TOTAL LANDED COST OPTIMIZER

- **Tech:** Python 3.14 + PuLP Optimization Engine
- **What it does:** Calculates total landed cost for all candidate ports in milliseconds
- **Physical checks:** Checks vessel draft vs port permissible depth.
- **Buffer check:** Ensures delivery before blast furnace stockpile runs dry.
- **Siding check:** Calculates train evacuation time (avoids port ground rent).

MODULE 3: ML FREIGHT FORECASTER

- **Tech:** XGBoost / Scikit-Learn Ensemble
- **What it does:** Forecasts 30-day forward freight rates (Baltic Dry Index - BDI).
- **Confidence:** Provides P10/P50/P90 predictive confidence intervals.
- **Market Drivers:** Tracks China iron ore demand & Australian cyclone weather.
- **Commercial Value:** Tells procurement officers whether to lock contracts or wait.

MODULE 4: 1-CLICK LEGAL & RAIL DOCS

- **Tech:** Automated Document Generation Engine
- **What it does:** Replaces days of lawyer emails with 1-click official documents.
- **BIMCO Notice:** Official Charter Party Rerouting Order for ship captain.
- **Master's Pre-NOR:** Official Notice of Readiness for destination port.
- **CRIS FOIS e-Indent:** Auto-books 39 BOXN railway rakes on Indian Railways.

1: THE 45-SECOND QUICK PITCH (MEMORIZ

"Respected judges, here is how OceanSteel AI works in 3 simple steps:

1. THE MAP & DATA: We built an interactive map using JavaScript and Leaflet. It displays live cargo ships
sailing in the Indian Ocean, port congestion on India's East Coast, and railway lines to steel plants.

2. THE PYTHON LOGIC: When a port like Paradip is jammed with a 7-day waiting queue, ships lose ₹20 Lakhs
every day in fines. Our Python code checks: Can this ship fit in a nearby port? Is the water deep enough? And are there train wagons ready to carry the coal inland?

3. THE SAVINGS: 10 days before the ship arrives, our system triggers an alert: 'Divert to Dhamra Port'.
Spending ₹3.5 Lakhs on extra fuel saves ₹1.4 Crores in waiting fines and ₹1 Crore on rail freight—saving ₹2.26 Crores and delivering raw material 8 days faster to the steel blast furnace!"

3-MINUTE PRESENTATION BREAKDOWN (MINUTE-BY-MINUTE)

MINUTE 1: THE PROBLEM & THE LOSS

- Explain that raw material logistics is 40% of steel production cost.
- Mention that Indian steel plants depend on imported Australian coking coal.
- State the crisis: Ships wait 7-10 days at Paradip, wasting $25,000/day (₹20 Lakhs/day) in Forex fines.
- The root cause: Sea tracking (AIS) and Indian Railways freight (FOIS) operate in complete silos.

MINUTE 2: THE LIVE DEMO & DECISION

- Show MV Steel Horizon 11 days out in the Indian Ocean carrying 150,000 MT of coal.
- Click 'Simulate Congestion': Paradip queue jumps to 7.8 days.
- Show our solver in action: Checks 17.5m draft, filters out shallow Haldia (8.5m), picks Dhamra.
- Point out the variance proof: Demurrage saved (167*k*)*heavilybeatsfuelcost*(4.2k) -> ₹2.26 Cr net saving!

MINUTE 3: OPERATIONAL EXECUTION & POLICY

- Click 'Generate Documents': Show auto-generated BIMCO Rerouting Addendum and CRIS FOIS e-Indent.
- Cite Calcutta High Court 2026 & Supreme Court 2020 rulings protecting the right to divert.
- Connect to National Logistics Policy: Lowers logistics cost to single digits and preserves Forex reserves.

ANSWER TRICKY QUESTIONS FROM

Q1: Did you write all this code yourself or did you use AI / vibe-code it?
"As first year students, we used AI coding tools to help us build the frontend and API structure, but the core logistics logic, the cost formulas, the railway rake math, and the problem architecture were designed entirely by us. We connected sea data and rail data into a working real-world prototype."

Q2: How is this different from MarineTraffic or VesselFinder?
"MarineTraffic is purely DESCRIPTIVE—it only tells you where a ship is. OceanSteel AI is PRESCRIPTIVE—it connects sea data to Indian Railways train schedules, calculates the cheapest alternate port, and tells the steel plant exactly what action to take to save money."

Q3: What if the ship is too large to fit in the new port?
"We implemented a strict physical water draft constraint. Giant Capesize ships sink 17.5m deep. Shallow riverine ports like Haldia (max draft 8.5m) are automatically blocked by our code. Only deep ports like Dhamra (18.5m) are allowed."

Q4: How do you know trains will be available at the new port?
"We track Indian Railways FOIS siding data. 150,000 MT of coal requires 39 BOXN rakes. Dhamra siding has 10 rakes/day ready, evacuating the cargo in 3.9 days (within free port days). If a port lacks trains, our code penalizes it with storage fines."

TRY JARGON & LEGAL PRECEDENTS (CHEAT SHEET)

- **Demurrage:** The penalty fine paid to foreign ship owners when a ship waits idle at port (~$25,000/day in USD).
- **Draft (Draught):** The vertical depth a vessel sits in water. Loaded ships need deep water (17.5m+).
- **Capesize:** Massive bulk ships (150,000+ tonnes) too big for Panama/Suez canals. Used for Australian coal.
- **BOXN Rake:** Standard 59-wagon open train used by Indian Railways to transport coal (~3,850 tonnes per train).
- **FOIS / CRIS:** Freight Operations Information System operated by Indian Railways to track freight trains.
- **BIMCO:** Baltic and International Maritime Council—world's largest body setting shipping contract standards.
- **Calcutta HC (2026):** SAIL vs. Vizag Seaport: Court affirmed charterers' right to divert ships to avoid port delays.
- **Supreme Court (2020):** Union of India vs. Inland Waterways: Enforced commercial duty to mitigate demurrage fines.